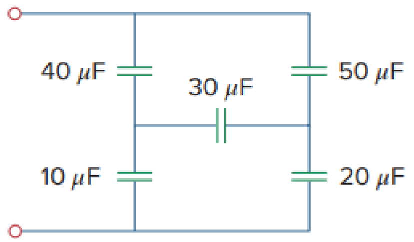
Chapter 6, Problem 28P

Problem

Obtain the equivalent capacitance of the network shown in c.

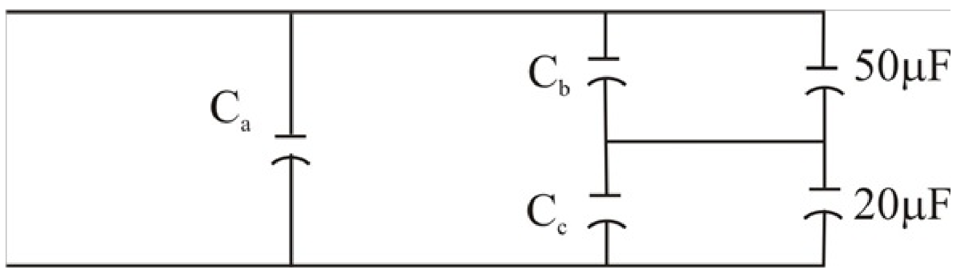
Fig. 6.60:



Step-by-step solution

Step 1 of 3

We may treat this like A resistive circuit and apply Delta – wye transformation, Except that R is replaced by 1/C



Step 2 of 3

$$\frac{1}{C_a}=\frac{\left(\frac{1}{10}\right)\left(\frac{1}{40}\right)+\left(\frac{1}{10}\right)\left(\frac{1}{30}\right)+\left(\frac{1}{30}\right)\left(\frac{1}{40}\right)}{\frac{1}{30}}$$

$$=\frac{3}{40}+\frac{1}{10}+\frac{1}{40}=\frac{2}{10}$$

$$C_a=5\mu F$$

$$\frac{1}{C_b}=\frac{\frac{1}{400}+\frac{1}{300}+\frac{1}{1200}}{\frac{1}{10}}=\frac{2}{30}$$

$$C_b=15\mu F$$

$$\frac{1}{C_c}=\frac{\frac{1}{400}+\frac{1}{300}+\frac{1}{1200}}{\frac{1}{40}}=\frac{4}{15}$$

Step 3 of 3

$C_c = 3.75\,\mu F$

C_b in parallel with $50\mu F = 50 + 15 = 65\,\mu F$

C_c in series with $20\mu F = 23.75\,\mu F$

65 in series with $23.75\,\mu F = \dfrac{65 \times 23.75}{88.75} = 17.39\,\mu F$

$17.39\,\mu F$ in parallel with $C_a = 17.39 + 5 = 22.39\,\mu F$

Hence $C_{eq} = 22.39\,\mu F$